

SHUBHAKARINI K

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PROFESSIONAL SUMMARY

Entry-level Software Engineer with strong skills in **Python, Artificial Intelligence, and application development**. Experienced in building AI-powered applications, automation tools, search engines, productivity software, and IoT solutions through academic projects and internships. Passionate about backend development, Generative AI, and designing efficient, scalable software solutions to solve real-world problems. Quick learner with strong problem-solving abilities and a continuous passion for learning new technologies.

TECHNICAL SKILLS

Programming Languages - Python - C - C++ Core Concepts - Object-Oriented Programming (OOP) - Data Structures - File Handling - Problem Solving	Artificial Intelligence - Prompt Engineering - Google Gemini API - Generative AI IoT & Embedded Systems - Arduino - Sensors - Embedded Systems Libraries - NumPy (Basics)	- Pandas (Basics) - Matplotlib (Basics) Tools & Platforms - Git - GitHub - VS Code - Arduino IDE - Tinkercad - Wokwi
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PROJECTS

AI-Powered Personalized Health Advisor (Generative AI)

- Developed an AI-powered healthcare assistant capable of understanding user queries and generating personalized health recommendations using Google Gemini API.
- Implemented prompt engineering techniques to provide context-aware and natural language responses.
- Designed a clean and interactive interface to improve user experience.
- Applied Generative AI concepts to build an intelligent conversational healthcare application.

Smart Appliance Usage Monitoring System with Auto Cutoff | Arduino | IoT

- Developed an IoT-based smart appliance monitoring system to improve electrical safety and energy efficiency.
- Monitored appliance usage through sensor data and automated relay-based power cutoff during unsafe conditions.
- Integrated sensors with Arduino for real-time monitoring and automation.
- Improved system reliability through embedded programming and hardware integration.

Mini Search Engine | Python

- Developed a lightweight search engine capable of indexing multiple text files and retrieving relevant results efficiently.
- Implemented keyword matching and organized search logic for faster information retrieval.
- Applied text processing and file handling concepts to manage indexed data.

URL Shortener | Python

- Built a URL shortening application to convert lengthy URLs into compact and shareable links.
- Implemented unique short-code generation and efficient URL mapping.
- Designed organized backend logic for storing and retrieving shortened URLs.
- Applied string manipulation and file handling techniques for efficient link management.

Code Complexity Analyzer | Python

- Developed a Python tool to analyze source code complexity and improve code quality.
- Implemented logical analysis to detect nested loops and conditional structures.
- Generated reports highlighting maintainability and readability concerns.
- Applied automation and file processing concepts for source code analysis.

Student Management System | Python

- Developed a console-based application to manage student records using CRUD operations.
- Implemented add, update, delete, and search functionalities through file handling.
- Designed modular program architecture for efficient data management.
- Improved application usability through organized backend logic.

EXPERIENCE

IoT Intern | RedMind Technologies
June 2025 – July 2025

- Developed and tested an IoT-based Smart Appliance Usage Monitoring System with Auto Cutoff.
- Worked with Arduino, sensors, relay modules, and embedded programming.
- Performed hardware testing, debugging, and real-time sensor integration.

Intern | Wearable MEMS Company
January 2026 – April 2026

- Assisted in AI- and sensor-based proposal development for engineering projects.
- Contributed to technical documentation, data analysis, project reporting, proposal preparation activities and collaborated with engineering teams during project development.

EDUCATION

Bachelor of Engineering (B.E.) - Electrical and Electronics Engineering
Government College of Engineering, Srirangam | 2022 – 2026 | CGPA: 7.82 / 10